



Agriculture: Corn Pest Management Guidelines

Common Smut (Boil Smut)

Ustilago maydis

Symptoms and Signs

Common smut is easily recognized by the tumorlike galls that form on any aboveground plant part. The conspicuous galls that replace kernels are covered with a greenish-white papery tissue. As the galls mature, the interior darkens and turns into masses of powdery, dark olive-brown to black spores. Ear galls may reach several inches in diameter. Galls that form on other plant parts, including the tassels and leaves, are much smaller.

Comments on the Disease

Spores overwinter in the soil. Under favorable conditions these form secondary spores that are carried by air currents or splashed by water to young, developing corn tissues. Development of common smut is favored by dry conditions and temperatures between 78° to 93°F. The incidence of smut is higher in soils high in nitrogen or after heavy applications of manure. Injury to the plant tissue of any kind increases the potential for smut infection.

Management

Although no corn variety is immune, some hybrids and varieties are more resistant than others. Ask your local farm advisor for varieties that perform well in your area. Avoid mechanical injuries to plants and maintain well-balanced soil fertility. Rotate to another crop, the longer the better.



UC IPM Pest Management Guidelines: Corn
UC ANR Publication 3443

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Text Updated: 01/06